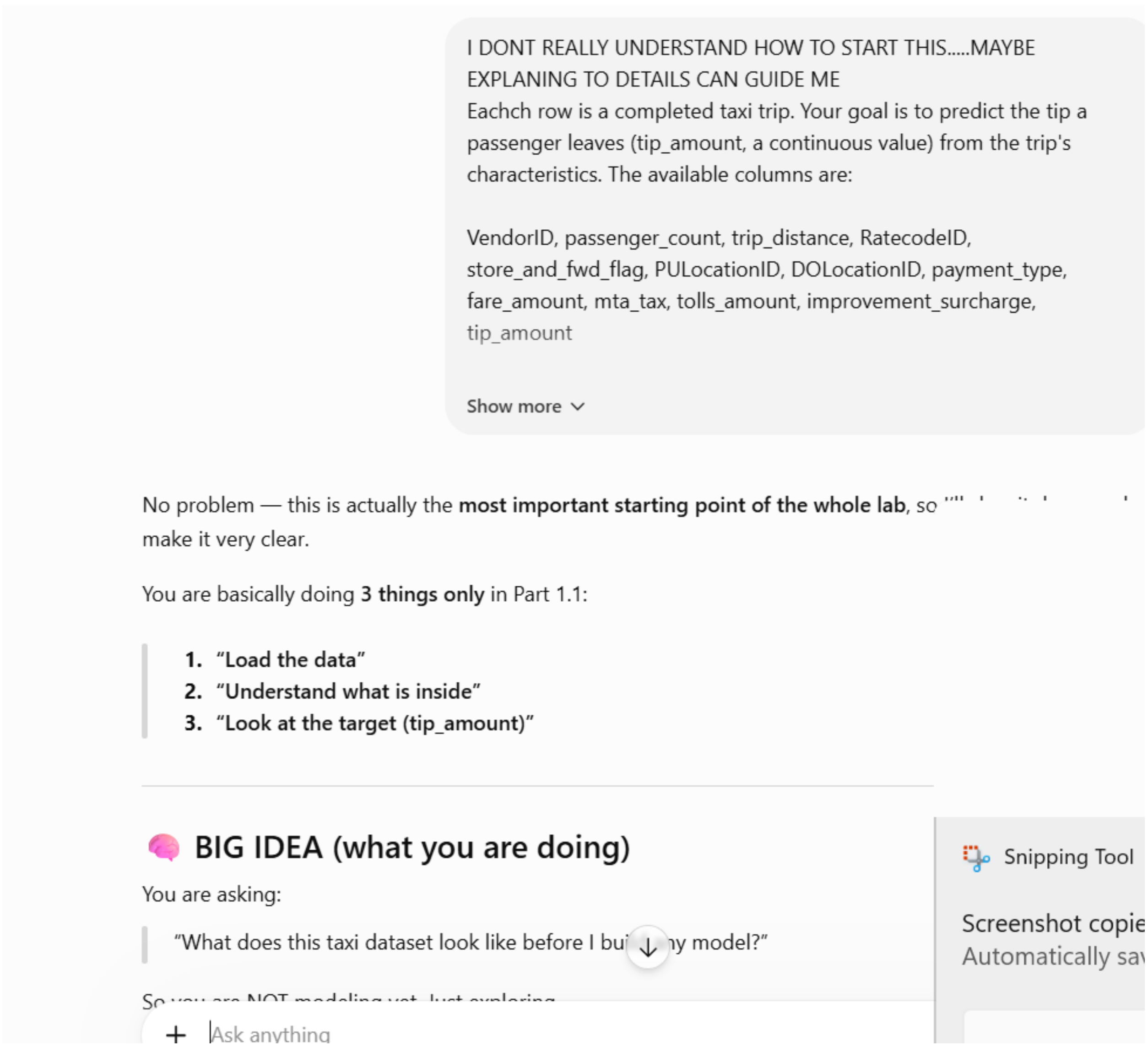
AI USED: CHAT GPT

PERCENTAGE: 20%

SCREENSHOTS ARE BELOW:



I DONT REALLY UNDERSTAND HOW TO START THIS.....MAYBE EXPLANING TO DETAILS CAN GUIDE ME
Eachch row is a completed taxi trip. Your goal is to predict the tip a passenger leaves (tip_amount, a continuous value) from the trip's characteristics. The available columns are:

VendorID, passenger_count, trip_distance, RatecodeID, store_and_fwd_flag, PULocationID, DOLocationID, payment_type, fare_amount, mta_tax, tolls_amount, improvement_surcharge, tip_amount

Show more

No problem — this is actually the **most important starting point of the whole lab**, so
make it very clear.

You are basically doing **3 things only** in Part 1.1:

> 1. **"Load the data"**
> 2. **"Understand what is inside"**
> 3. **"Look at the target (tip_amount)"**

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## 🧠 BIG IDEA (what you are doing)

You are asking:

> "What does this taxi dataset look like before I bui ny model?"

Ask anything

EXPLAIN TO ME WHAT IAM SUPPOSED TO DO PLS
TODO: Handle missing / invalid rows
# e.g. drop or impute NaNs; remove rows where trip_distance <= 0 c
fare_amount <= 0;
# decide how to treat negative tip_amount values.

# TODO: Feature engineering — create at least ONE new, justified feature, for example:
# - tip_rate proxy, fare-per-mile = fare_amount / trip_distance
# - total surcharges = mta_tax + tolls_amount + improvement_surcharge

Show more

Good — this is the part where most people get confused. I'll break it down into **very simple actions** so yc know exactly what to do.

Think of this step as:

> "Clean the taxi data so a machine learning model can understand it."

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## 🧠 PART 1 — Handle missing + invalid data

### ❓ What are you doing?

You are removing "bad rows" that don't make sense.

#### 🚫 Examples of bad data:

- missing values (NaN)
- trip_distance ≤ 0 (impossible trip)

Ask anything

EXPLAIN THE CODE VERY WELL ....LAYMAN TERMS

Alright — I'll explain it in **very simple, everyday language**, like you've never seen machine learning before.

Think of this whole section as:

> 🧹 "Cleaning and preparing taxi data so a computer can learn from it."

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## 🧠 1. Handling missing + invalid data

**Code:**

Python

Run

```python
taxi = taxi.dropna()
taxi = taxi[taxi["trip_distance"] > 0]
taxi = taxi[taxi["fare_amount"] > 0]
taxi = taxi[taxi["tip_amount"] >= 0]
```

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### 💡 What this REALLY means:

Imagine your taxi data is like a notebook filled with trip records.

Some pages are:

- blank (missing data)
- nonsense (trip distance = 0 → no trip happened)

Ask anything

ChatGPT can make mistakes. Check important info.

## 📊 FINAL SPLIT

You now have:

- 60% → Train
- 20% → Validation
- 20% → Test

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## ⚖️ STEP 4 — Scaling (VERY IMPORTANT RULE)

Python

```python
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)  # Learn from train ONLY
X_val_scaled = scaler.transform(X_val)
X_test_scaled = scaler.transform(X_test)
```

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## 🧠 WHY "FIT ONLY ON TRAIN"?

Simple meaning:

> You must not let the model "see" validation or test data during learning.

Otherwise it's like:
👉 studying with exam answers already visible

Ask anything

WHAT IS IT ABOUT
I WANT TO KNOW AND UNDERSTAND WHAT TO DO HERE
TODO: Use ONLY the scaled obesity features (no target) for clustering.

# TODO: Choose the number of clusters k with the Elbow method (and/or silhouette score).
#   from sklearn.cluster import KMeans
#   from sklearn.metrics import silhouette_score
#   Loop k = 2..10, record inertia_ (and silhouette), and plot inertia vs k.

# TODO: Fit KMeans with your chosen k

Show more

Good — this is the **core idea of your unsupervised learning section**, so I'll explain it in plain, simple terms first.

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## 🧠 WHAT THIS SECTION IS ABOUT (BIG IDEA)

You are pretending:

> "I DO NOT KNOW the obesity labels (NObeyesdad)."

Then you ask:

> "Can the computer still find natural groups of people just from their features?"

That's it.

This is called:

Ask anything

I DID SOMETHING BUT IT IS NOT WORKING AND IT DOESNT REALLY MAKE SENSE....COULD U GIVE ME YOURS SO I CAN RECTIFY MY MISTAKE

ot you — I'll give you a **clean working version** of the whole K-Means section so you can compare and fix ours.

I also keep it simple so you see exactly what each part is doing.

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## ✅ 🧩 FULL WORKING K-MEANS PIPELINE (COPY THIS)

### . Use ONLY features (remove target)

Python

```python
X_cluster = obesity_clean.drop("NObeyesdad", axis=1)
```

---

### . Find best k (Elbow + Silhouette)

Python

```python
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
```

Ask anything

YOUR OWN IS VERY DIFFERENT FROM MINE.....EXPLAIN IT CODE BY CODE

Good — let's slow this right down and go **line by line exactly what each piece of code is doing**, in plain language. Forget "ML theory" for now — think of it like following a cooking recipe.

I'll also show you what each line *means in real life*.

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## 🧩 STEP 1 — Remove the answer column

Python

Run

```python
X_cluster = obesity_clean.drop("NObeyesdad", axis=1)
```

### 🧠 What this means:

You are saying:

> "I want ONLY the input information, not the answers."

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**Think of it like:**

You give a student data:

- Age

Ask anything

## 🧩 STEP 5 — Build KMeans model

Python

```python
kmeans = KMeans(n_clusters=k, random_state=RANDOM_STATE, n_init=10)
```

### 🧠 Meaning:

You are saying:

> "Split data into k groups"

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**Break it down:**

- `n_clusters=k` → number of groups
- `random_state` → ensures same results every time
- `n_init=10` → tries multiple starts for better grouping

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**Think of it like:**

You are asking:

> "Split these people into k groups and try 10 different ways to do it, then pick the best one."

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## 🧩 STEP 6 — Assign clusters

Python

Ask anything